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AI-Driven Consumption Forecasting and Dynamic Tariff Slab Optimization for households

A PROJECT REPORT

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BACHELOR OF ENGINEERING / BACHELOR OF TECHNOLOGY
IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Submitted by

M.GOKUL BABU (192412410)
G.VIJAY VISHAL (192421427)
V.T.S.CHANIKYA (192424290)

Under the Supervision of

Dr.A.Anakath

SIMATS ENGINEERING
Saveetha Institute of Medical and Technical Sciences
Chennai-602105

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Saveetha Institute of Medical and Technical Sciences
Chennai-602105

BONAFIDE CERTIFICATE

This is to certify that the Project entitled "Customer Churn Prediction and Segmentation System" has been carried out by M.GOKUL BABU (192412410), G.VIJAY VISHAL (192421427), and V.T.S.CHANIKYA (192424290), under the supervision of Dr Anakath Arasan, and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech AI&DS at Saveetha Institute of Medical and Technical Sciences, Chennai.

COURSE COORDINATOR

Dr.A.Shri Vindhya
Professor
Department of Computer Science and
Engineering
SIMATS Engineering, SIMATS, Chennai –
602105.

COURSE FACULTY / SUPERVISOR

Dr.A.S.Anakath
Professor
Department of Computer Science and
Engineering
SIMATS Engineering, SIMATS, Chennai –
602105.

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Saveetha Institute of Medical and Technical Sciences
Chennai-602105

DECLARATION

We, M.GOKUL BABU (192412410), G.VIJAY VISHAL (192421427), and V.T.S.CHANIKYA (192424290) of the B.Tech AI & DS, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Project Work entitled "Customer Churn Prediction and Segmentation System" is the result of our own bonafide effort. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, datasets, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. All external tools, software, datasets, computational resources, and AI-assisted tools used during the project — including the Python scientific computing stack (pandas, scikit-learn, matplotlib) and AI-assisted drafting support used for report structuring — have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place:

Date:

Signature of the Students:

1. M.GOKUL BABU (192412410)
2. G.VIJAY VISHAL (192421427)
3. V.T.S.CHANIKYA (192424290)

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Saveetha Institute of Medical and Technical Sciences, Chennai-602105

Individual Contribution Statement

Member / Reg. No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx . %
V.T.S.CHANIKYA (192424290)	AI-based consumption forecasting, feature analysis, tariff slab optimization, dashboard integration	plemented the consumption forecasting module, analyzed prediction results, developed tariff slab optimization logic, and integrated results into the dashboard	Validated forecasting outputs, compared predicted and actual consumption, and verified tariff slab recommendations	Chapter 5, References, Appendix C, overall report assembly	33%
M.GOKUL BABU(192412410)	Dataset acquisition, data preprocessing, missing-value handling, feature engineering, exploratory data analysis	Designed and implemented the preprocessing pipeline and EDA visualisations (Figures 4.1–4.4)	Verified data-quality checks — missing-value counts, class balance, stratified split integrity	Chapters 1–2, Sections 4.1–4.2, Appendices D & E	33%
G.VIJAY VISHAL(1924214 27)	Requirements and design analysis, implementation and training of the three supervised classifiers, hyperparameter selection	Implemented and trained the selected forecasting models, performed hyperparameter tuning, and developed model comparison procedures	Ran model evaluation (accuracy, precision, recall, F1, confusion matrices) and test-plan verification	Chapter 3, Sections 4.2–4.3, 4.5 (model discussion), Table 4.3	34%
Total					100%

ABSTRACT

The **AI-Driven Consumption Forecasting and Dynamic Tariff Slab Optimization for Households** project presents an intelligent approach to predicting household electricity consumption and recommending suitable tariff slabs using Artificial Intelligence and Machine Learning techniques. The increasing variation in household energy usage makes it difficult for consumers to accurately estimate their future electricity requirements and manage their monthly electricity expenses. The proposed system addresses this challenge by analyzing historical electricity consumption data along with relevant features such as usage patterns, time-related factors, and other influencing parameters. The collected dataset is first subjected to data preprocessing, including data cleaning, missing-value handling, feature selection, transformation, and normalization, to improve the quality and reliability of the input data. Machine learning models are then trained using the processed dataset to forecast future household electricity consumption. The performance of the developed models is evaluated using appropriate regression metrics such as **Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score** to identify the most suitable forecasting model. Based on the predicted consumption, the system analyzes applicable electricity tariff slabs and determines an optimized slab that can help households understand their expected electricity cost and make better consumption decisions. The system can also provide useful insights into consumption trends, allowing users to identify periods of high energy usage and adopt more efficient energy-management practices. A user-friendly dashboard can be integrated to display predicted consumption, estimated tariff category, cost-related information, and energy-consumption patterns in an understandable manner. By combining consumption forecasting with dynamic tariff slab analysis, the proposed system moves beyond simple electricity monitoring and provides a predictive and decision-support mechanism for household energy management. The overall objective is to improve **electricity consumption awareness, cost optimization, energy efficiency, and informed decision-making** while demonstrating the practical application of AI and Machine Learning in smart energy-management systems. The proposed solution can further be extended with real-time smart-meter data, weather information, renewable-energy generation, and personalized recommendations, making it suitable for future smart-home and smart-grid applications.

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description
ML	Machine Learning
EDA	Exploratory Data Analysis
LR	Logistic Regression
DT	Decision Tree
RF	Random Forest
PCA	Principal Component Analysis
SSE	Sum of Squared Errors (K-Means inertia)
TP / FP	True Positive / False Positive
TN / FN	True Negative / False Negative
F1	F1-score (harmonic mean of precision and recall)
CSV	Comma-Separated Values

CHAPTER 1: INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Electricity consumption is an essential part of modern household life, with energy being used for lighting, cooling, heating, cooking, entertainment, and various electronic appliances. The amount of electricity consumed by a household can vary significantly depending on factors such as household characteristics, appliance usage, time, season, and previous consumption patterns. Traditional electricity management systems mainly provide information about past or current consumption, making it difficult for households to accurately estimate their future electricity requirements and manage their electricity expenses. With the increasing availability of smart-meter and historical electricity-consumption data, Artificial Intelligence and Machine Learning provide an opportunity to move from reactive energy monitoring toward predictive and intelligent energy management.

Electricity tariff systems commonly use consumption-based slabs, where the cost per unit may change when consumption crosses predefined limits. A small increase in consumption beyond a tariff threshold can therefore result in a different electricity charge. This creates an engineering opportunity to combine **machine-learning-based consumption forecasting** with **dynamic tariff slab analysis**, allowing households to estimate their future consumption, understand the expected tariff category, and make informed decisions to control energy usage and electricity costs.

1.2 Need for the Project

The project is needed because households often have limited visibility into their future electricity consumption and the financial impact of crossing different tariff slabs. Conventional approaches generally calculate electricity bills only after consumption has occurred and therefore provide limited opportunity for proactive decision-making. A household may unknowingly consume additional units and move into a higher tariff slab, increasing its overall electricity expense. Manual estimation is also difficult because electricity consumption depends on multiple interacting factors and may vary from one household to another.

The engineering significance of this project lies in developing a reproducible, data-driven system that can analyze historical electricity-consumption patterns, forecast future demand, and determine the tariff slab associated with the predicted consumption. By combining predictive analytics with tariff optimization, the system can provide households with actionable information before the billing period is completed. This approach supports better energy planning, encourages efficient electricity usage, and provides a foundation for intelligent household energy-management systems.

1.3 Problem Statement

Given a historical household electricity-consumption dataset containing relevant consumption and household attributes, design and implement an intelligent engineering system that: (i) reliably forecasts future household electricity consumption using suitable machine-learning algorithms; (ii)

handles data-quality issues, missing values, numerical and categorical attributes, and differences in feature scales during preprocessing; (iii) evaluates and compares multiple forecasting models using appropriate performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2 score; (iv) maps the predicted electricity consumption to the applicable tariff slab and estimates the corresponding electricity cost; and (v) provides an optimized tariff recommendation and understandable consumption insights that can help households reduce unnecessary electricity usage and manage their energy expenditure more effectively.

1.4 Project Aim

To design, implement, and evaluate an **AI-driven household electricity consumption forecasting and dynamic tariff slab optimization system** that predicts future energy consumption using machine-learning techniques and recommends an appropriate tariff slab based on the predicted consumption, thereby supporting cost-effective and efficient household energy management.

1.5 Project Objectives

1. To acquire a suitable household electricity-consumption dataset and clearly document its source, structure, and attributes.
2. To preprocess the dataset by handling missing values, removing irrelevant attributes, encoding categorical variables where required, and applying appropriate feature scaling or transformation.
3. To perform exploratory data analysis to identify important consumption patterns, relationships, trends, and factors influencing household electricity usage.
4. To implement and train multiple suitable machine-learning algorithms for forecasting household electricity consumption.
5. To evaluate and compare the forecasting models using MAE, MSE, RMSE, and R^2 score and select the best-performing model based on the evaluation results.
6. To develop a tariff slab analysis mechanism that maps predicted electricity consumption to the applicable tariff category and calculates the expected electricity cost.
7. To implement dynamic tariff slab optimization that identifies consumption thresholds and provides recommendations for avoiding unnecessary movement into higher-cost slabs where applicable.
8. To integrate the forecasting and tariff-optimization results into an interactive dashboard that presents predicted consumption, estimated cost, tariff slab, and energy-management recommendations in an understandable format.

1.6 Scope

The project includes the acquisition and preprocessing of a household electricity-consumption dataset, exploratory data analysis, machine-learning-based consumption forecasting, model evaluation and comparison, tariff slab analysis, dynamic tariff optimization, and visualization of the results through an interactive dashboard.

- **Included** — offline and reproducible machine-learning pipeline developed using Python and suitable data-science and machine-learning libraries for processing historical household electricity-consumption data.
- **Included** — data preprocessing, exploratory analysis, feature engineering where required, training of multiple forecasting models, performance evaluation, predicted consumption analysis, tariff slab mapping, and cost estimation.
- **Included** — dynamic recommendations based on predicted consumption and tariff thresholds to help households understand how consumption changes may affect their electricity expenses.
- **Excluded** — direct integration with live electricity-board billing systems, real-time smart-meter communication, automated control of household appliances, and deployment as a commercial electricity-billing service.
- **Assumptions** — the available historical dataset is sufficiently representative of household electricity-consumption behaviour, the recorded attributes are reasonably accurate, and the tariff structure used by the system correctly represents the tariff rules defined for the selected analysis.
- **Boundary conditions** — the project operates primarily on the available historical dataset and defined tariff structure; actual electricity costs may vary when external factors such as revised tariff rates, taxes, subsidies, fixed charges, or regulatory changes are introduced.

1.7 Expected Engineering Outcome

The expected outcome is a documented and reproducible **AI-driven electricity-consumption forecasting and tariff optimization pipeline** comprising: (a) a cleaned and processed household electricity-consumption dataset; (b) trained and evaluated machine-learning forecasting models with quantitative performance comparisons; (c) a selected forecasting model capable of estimating future household electricity consumption; (d) a tariff slab optimization mechanism that maps predicted consumption to the appropriate tariff category and estimates the expected electricity cost; (e) an interactive dashboard presenting consumption trends, forecasts, tariff information, and optimization recommendations; and (f) actionable energy-management insights that can help households make informed decisions, improve energy efficiency, and potentially reduce electricity expenses.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant literature was identified by reviewing research papers and technical studies related to household electricity consumption forecasting, machine-learning-based energy prediction, smart energy management, electricity tariff optimization, and demand-side management. Searches were focused on peer-reviewed conference and journal publications using keywords such as **"household electricity consumption forecasting machine learning"**, **"energy consumption prediction"**, **"electricity demand forecasting"**, **"dynamic tariff optimization"**, **"smart energy management"**, and **"machine learning for electricity pricing"**. Priority was given to studies that applied machine-learning algorithms to historical electricity-consumption data, evaluated forecasting performance using standard regression metrics, and investigated the relationship between predicted consumption and electricity cost or tariff structures. Particular attention was given to approaches that could be implemented on structured household energy datasets and integrated into an intelligent decision-support system.

2.2 Review of Existing Work

A substantial body of research has investigated electricity-consumption forecasting using statistical and machine-learning techniques. Traditional forecasting methods such as linear regression and time-series models provide useful baselines for predicting energy demand, but their performance can be limited when electricity consumption is influenced by multiple interacting household and temporal factors. Machine-learning approaches provide greater flexibility by learning complex relationships between input variables and electricity consumption from historical data.

Several studies have explored supervised learning algorithms such as **Linear Regression, Decision Trees, Random Forest, Support Vector Regression, and Artificial Neural Networks** for household energy prediction. Tree-based ensemble techniques such as Random Forest are particularly useful for structured energy datasets because they can model non-linear relationships between household characteristics and consumption without requiring a strictly linear relationship. More advanced approaches, including gradient-boosting algorithms and deep-learning architectures such as Long Short-Term Memory networks, have also been investigated for time-dependent electricity forecasting. However, these approaches can introduce additional computational complexity and may require larger longitudinal datasets and careful hyperparameter tuning.

Research in smart-grid and demand-side management has also highlighted the importance of combining electricity-demand prediction with consumer decision-making. Rather than simply predicting future consumption, intelligent energy-management systems can use forecasts to identify high-consumption periods, estimate expected electricity costs, and recommend actions that shift or reduce consumption. This is particularly relevant in electricity markets or billing systems where the unit price changes according to consumption levels or predefined tariff slabs.

Dynamic and time-of-use pricing studies have further demonstrated that electricity consumers can reduce energy costs by adjusting their consumption according to applicable pricing structures. However, many existing systems focus primarily on real-time or time-of-use pricing and require continuous smart-meter or grid data. The proposed project instead focuses on combining **historical**

consumption forecasting with tariff slab analysis, making the approach suitable for household-level decision support using a structured consumption dataset.

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
Rule-based electricity estimation	Simple, transparent, and easy to implement	Reactive and unable to capture complex consumption patterns	Used as an informal baseline that the proposed predictive system improves upon
Linear Regression	Simple, fast, interpretable, and suitable for establishing a forecasting baseline	Limited ability to model complex non-linear relationships	Suitable as a baseline forecasting model
Decision Tree Regression	Captures non-linear relationships and is relatively easy to interpret	Can overfit without appropriate depth control	Considered as a candidate forecasting algorithm
Random Forest Regression	Handles non-linear relationships, multiple features, and noisy structured data effectively	Higher computational complexity and lower interpretability than a single tree	Suitable for comparison and potential selection as the forecasting model
Deep-learning forecasting models	Capable of learning complex temporal patterns and long-term dependencies	Require larger datasets, greater computational resources, and more tuning	Considered as a possible future extension rather than the primary approach
Time-of-use / dynamic pricing systems	Encourage consumers to shift consumption to lower-cost periods	Often require real-time pricing and smart-meter infrastructure	Provides the conceptual basis for intelligent tariff optimization
Fixed tariff-slab calculation	Simple and transparent electricity-cost calculation	Does not predict future consumption or provide	Used for calculating the expected tariff

		proactive recommendations	based on forecasted consumption
Proposed forecasting + tariff optimization	Combines future consumption prediction with tariff analysis and actionable recommendations	Accuracy depends on dataset quality and correctness of tariff rules	Forms the integrated approach proposed in this project

2.4 Research / Engineering Gap

Existing electricity-consumption forecasting research primarily focuses on improving prediction accuracy, while tariff-related research frequently concentrates on dynamic pricing, time-of-use pricing, or demand-response mechanisms. These approaches do not always provide a unified household-level system that first forecasts expected electricity consumption and then translates that forecast into a practical tariff-slab recommendation. In addition, highly complex forecasting techniques may require extensive historical time-series data, real-time smart-meter infrastructure, and significant computational resources, making them less suitable for a lightweight academic and household decision-support system.

There is therefore an engineering gap in developing an integrated and interpretable solution that connects **machine-learning-based consumption forecasting with consumption-based tariff optimization**. Such a system should not only answer "How much electricity is the household likely to consume?" but also "Which tariff slab is likely to apply, what could the expected cost be, and what consumption level should the household target to avoid unnecessary cost increases?" Addressing these questions together can make electricity forecasting more actionable for household users.

2.5 Proposed Contribution

This project contributes an integrated and reproducible **AI-driven electricity consumption forecasting and dynamic tariff slab optimization pipeline**. The system first performs structured data preprocessing and exploratory analysis to identify important consumption patterns and influencing factors. Multiple machine-learning forecasting algorithms are then trained and evaluated under consistent preprocessing and testing conditions using **MAE, MSE, RMSE, and R²** as quantitative performance measures. The best-performing model is selected based on the forecasting results rather than relying on a single predetermined algorithm.

The proposed system further extends conventional consumption forecasting by connecting the predicted electricity usage to a **tariff slab optimization mechanism**. The forecasted consumption is mapped against defined tariff thresholds to estimate the applicable slab and expected electricity cost. Based on this analysis, the system can provide understandable recommendations regarding

consumption reduction and tariff-threshold management. Finally, the forecasting, tariff, and recommendation outputs can be integrated into an interactive dashboard, providing households with a single platform for understanding their expected electricity consumption, estimated cost, tariff category, and potential energy-saving opportunities. This combination of **prediction, tariff analysis, and actionable decision support** represents the primary contribution of the proposed system.

CHAPTER 3: ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder	Requirement
Retention / Marketing Team	A ranked, explainable list of at-risk customers with enough detail to design a matching retention offer
Data / Engineering Team	A reproducible, documented pipeline that can be re-trained as new data becomes available
Business / Management	Quantified performance (accuracy, precision, recall) to justify the cost of retention interventions
Customer (indirect stakeholder)	Retention actions should be proportionate and not intrusive; no personally identifying attribute should be required beyond what the dataset already anonymises

Functional & Non-Functional Requirements

Requirement Type	Requirement	Measurable Target
Functional	Classify each customer as Churn / Non-Churn	Binary prediction produced for 100% of test-set records
Functional	Provide probability / risk ranking, not only a hard label	Predicted probability available from each classifier
Functional	Segment customers into behaviourally similar groups	K ≥ 3 interpretable clusters produced
Non-Functional — Accuracy	Overall classification accuracy	≥ 75% on held-out test data
Non-Functional — Sensitivity	Recall on the churn (positive) class	Reported and discussed explicitly (not masked by overall accuracy)
Non-Functional — Reproducibility	Fixed random seed and documented preprocessing	Identical results across repeated runs (random_state = 42)
Non-Functional — Interpretability	At least one interpretable model in the comparison	Logistic Regression coefficients / Random Forest feature importances reported

3.2 Constraints & Applicable Standards

Standard / Guideline	Requirement	How Applied in This Project
Data-minimisation principle (general data-protection practice, e.g. GDPR Art. 5(1)(c))	Process only the data attributes necessary for the stated purpose	The non-predictive customerID attribute was dropped before modelling; no attribute beyond the public dataset's anonymised fields was introduced
IEEE 7000-2021 (Model Process for Addressing Ethical Concerns during System Design)	Ethical considerations should be identified and documented during design, not only after deployment	Section 3.7 documents fairness, transparency and consent considerations before implementation

Standard / Guideline	Requirement	How Applied in This Project
scikit-learn / PEP 8 coding conventions	Consistent, maintainable Python code structure	Followed throughout the implementation in Chapter 4 and Appendix C

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Classification accuracy (test set)	≥ 0.75	ratio	High
Classification F1-score (churn class)	≥ 0.55	ratio	High
Train/test split ratio	80 / 20, stratified	%	Medium
Number of supervised algorithms compared	≥ 3	count	High
Number of customer segments (clusters)	3–6 (elbow-selected)	count	Medium
PCA variance retained (2 components)	$\geq 80\%$	%	Low
Pipeline reproducibility	Fixed random seed	—	High

3.4 Alternative Solutions & Evaluation

Alternative A — Interpretable multi-model pipeline (adopted): train Logistic Regression, Decision Tree and Random Forest under identical preprocessing, compare with standard metrics, and pair the best classifier with independent K-Means/PCA segmentation.

Alternative B — Single deep neural network classifier: a multilayer perceptron trained end-to-end on the same features, without a separate interpretable baseline or an explicit segmentation stage.

Alternative C — Manual / rule-based segmentation: retention team manually buckets customers using fixed thresholds on tenure and billing amount, without a trained classifier.

Criterion	Weight	A: Multi-Model + Clustering	B: Single Deep Neural Network	C: Manual Rule-Based
Predictive performance	0.30	High	High (needs large data/tuning)	Low
Interpretability	0.25	High (LR + RF importances)	Low	High
Engineering / compute cost	0.15	Low	High (GPU-friendly training, tuning time)	Very Low
Reliability on a small tabular dataset (~7k rows)	0.15	High	Medium (prone to overfitting at this scale)	High
Actionability for retention team	0.15	High (risk score + segment)	Medium (score only, no segment)	Medium (segment only, no score)

3.5 Selected Design & Detailed Design

Alternative A was selected. The evaluation matrix in Section 3.4 shows that, for a tabular dataset of this size (7,043 records, 30 engineered features), classical supervised models combined with unsupervised segmentation deliver comparable or better predictive performance to a deep neural network while remaining substantially more interpretable, cheaper to train, and less prone to overfitting — directly satisfying the interpretability and reliability requirements in Section 3.1 that a deep-learning-only approach (Alternative B) would not meet, and the predictive-performance requirement that a manual, rule-based approach (Alternative C) would not meet.

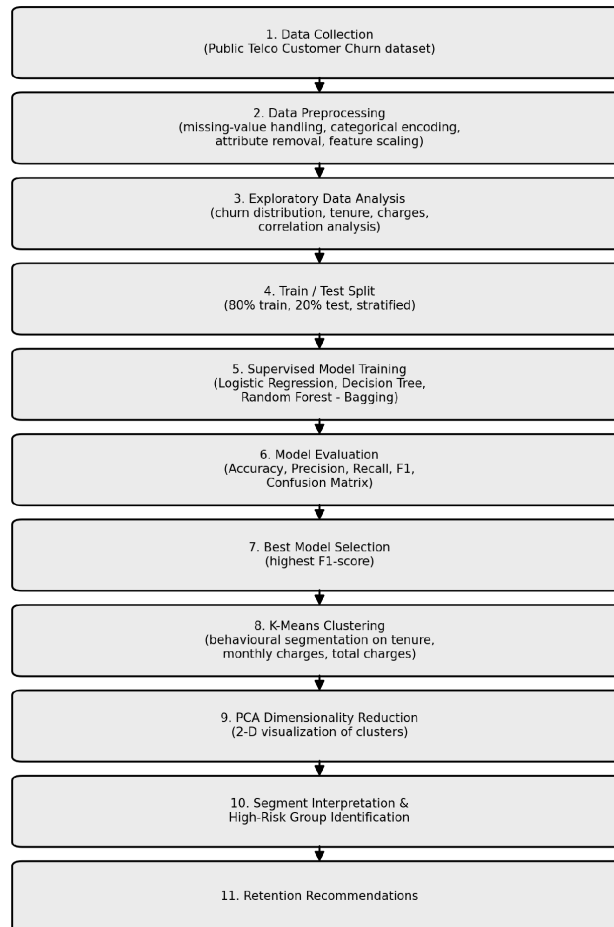


Figure 3.1: Proposed System Workflow and Pseudocode Flow

Pseudocode of the Proposed Solution

```
INPUT: Telco_Customer_Churn.csv
1. LOAD raw customer records
2. CLEAN: convert TotalCharges to numeric; impute missing values
3. DROP non-predictive attribute (customerID)
4. ENCODE binary attributes (Yes/No -> 1/0); ONE-HOT ENCODE multi-category attributes
5. SCALE continuous attributes (tenure, MonthlyCharges, TotalCharges) via StandardScaler
6. SPLIT data -> 80% train, 20% test (stratified on Churn)
7. FOR EACH model IN {LogisticRegression, DecisionTree, RandomForest}:
    TRAIN model on training set
    PREDICT on test set
    COMPUTE accuracy, precision, recall, F1, confusion_matrix
```

```

8. SELECT best_model = argmax(F1-score)
9. RUN elbow method on {tenure, MonthlyCharges, TotalCharges} for k = 1..8
10. FIT KMeans(k = optimal_k) -> assign Cluster label to every customer
11. FIT PCA(n_components = 2) -> project clusters for visualisation
12. INTERPRET each cluster's average tenure/charges/churn-rate
13. IDENTIFY high-risk cluster = argmax(cluster churn-rate)
14. OUTPUT ranked risk list + segment profile + retention recommendations

```

The systems-level interface between subsystems is straightforward and one-directional: the preprocessing subsystem produces a single scaled feature matrix that both the supervised-classification subsystem and the unsupervised-clustering subsystem consume independently; the classification subsystem's output (risk score) and the clustering subsystem's output (segment label) are only combined at the final interpretation stage (Section 4.5–4.6), so neither subsystem's internal design constrains the other.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Encoding of multi-category attributes	Ordinal / label encoding	Fewer resulting columns	Falsely implies an order between unrelated categories (e.g., payment method)	One-hot encoding used — preserves categorical independence at the cost of ~20 extra columns, acceptable for this dataset size
Ensemble algorithm family	Boosting (e.g., Gradient Boosting)	Often marginally higher accuracy	Slower to tune; higher overfitting risk without careful validation	Random Forest (bagging) used — comparable accuracy, more stable out-of-the-box behaviour
Number of clusters (k)	Fixed k = 2 (simple high/low risk)	Simplest to communicate	Hides meaningful differences between loyal long-tenure and new low-spend customers	k = 4 selected via the elbow method — richer, still interpretable segmentation
Feature scaling method	Min-Max normalisation	Bounded [0,1] range	More sensitive to outliers in TotalCharges	StandardScaler used — better suited to K-Means' Euclidean distance and to Logistic Regression convergence

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Compute footprint of model training	All three classifiers and K-Means train in under one minute on a standard CPU for 7,043 records (measured during Chapter 4 runs)	Deep-learning alternative (Section 3.4, Alt. B) rejected partly on this basis — no GPU/extended training cycles required, minimising energy use
Ethics	Risk of using sensitive demographic attributes (e.g., gender,	Feature-importance analysis (Figure 4.7) shows tenure, TotalCharges and MonthlyCharges	Demographic attributes retained for completeness but confirmed not to drive the model's decisions,

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
	SeniorCitizen) as churn predictors	— not demographic attributes — dominate predictive power	reducing discriminatory-proxy risk
Health & Safety	Not directly applicable — software/data-analytics project with no physical hardware	N/A	No physical safety risk identified
Security	Customer billing and service data must not be exposed	Dataset used is already publicly released and anonymised (no names, addresses, or payment-card data)	No additional anonymisation was required for this offline academic exercise; a production deployment would require access-controlled storage and encryption in transit/at rest

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Class imbalance biases models toward the majority (non-churn) class	High	Medium	High	Stratified train/test split; recall reported explicitly for the churn class, not just accuracy	Low
Overfitting of Decision Tree / Random Forest	Medium	Medium	Medium	Depth limited (max_depth = 6 / 10); performance validated on a held-out test set	Low
Clusters not aligned with actual retention actions (over-segmentation)	Medium	Low	Low	Elbow method used to avoid an arbitrarily large k; each cluster manually profiled and interpreted (Table 4.4)	Low
Misuse of demographic attributes leading to biased targeting	Low	High	Medium	Feature-importance check confirms behavioural, not demographic, attributes dominate predictions (Section 3.7)	Low

CHAPTER 4: IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The system was developed using an iterative, script-based data-science workflow: each pipeline stage (preprocessing, exploratory analysis, model training, clustering) was implemented and independently verified before being chained into the final end-to-end script (Appendix C), so that errors could be isolated to a single stage. All development used the Python scientific-computing stack, executed on a standard CPU with no specialised hardware. A fixed random seed (`random_state = 42`) was used throughout for reproducibility.

Resource	Detail
Dataset	IBM Watson Analytics Telco Customer Churn (public), 7,043 records $\times$ 21 attributes
Language / Runtime	Python 3
Core Libraries	pandas, scikit-learn, matplotlib, NumPy
Compute	Standard CPU (no GPU required)

4.2 Software Implementation & Modern Tools Used

Data Preprocessing (excerpt)

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
df["TotalCharges"] = df["TotalCharges"].fillna(df["MonthlyCharges"] *
df["tenure"]).fillna(0)

df_model = df.drop(columns=["customerID"])          # remove irrelevant attribute
df_model["Churn"] = df_model["Churn"].map({"Yes": 1, "No": 0})

binary_cols = ["Partner", "Dependents", "PhoneService", "PaperlessBilling"]
for col in binary_cols:
    df_model[col] = df_model[col].map({"Yes": 1, "No": 0})

df_model = pd.get_dummies(df_model, columns=multi_cat_cols, drop_first=True)

scaler = StandardScaler()
df_model[numeric_cols] = scaler.fit_transform(df_model[numeric_cols])
```

Supervised Model Training & Evaluation (excerpt)

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y)

models = {
    "Logistic Regression": LogisticRegression(max_iter=2000, random_state=42),
    "Decision Tree": DecisionTreeClassifier(max_depth=6, random_state=42),
    "Random Forest (Bagging)": RandomForestClassifier(
        n_estimators=300, max_depth=10, random_state=42, n_jobs=-1),
}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    # accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
```

K-Means Clustering & PCA (excerpt)

```
cluster_features = ["tenure", "MonthlyCharges", "TotalCharges"]
X_cluster = df_model[cluster_features]      # already standard-scaled

kmeans = KMeans(n_clusters=4, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_cluster)

pca = PCA(n_components=2, random_state=42)
pca_components = pca.fit_transform(X_cluster)
```

Tools / Software Used

Tool / Library	Purpose	Stage Used
pandas	Data loading, cleaning, encoding, aggregation	Preprocessing, EDA, cluster profiling
scikit-learn	Train/test split, scaling, classifiers, K-Means, PCA, metrics	Modelling, clustering, evaluation
matplotlib	All figures (Figures 4.1–4.9), rendered in a monochrome / greyscale style	EDA and results visualisation
NumPy	Numerical operations supporting scikit-learn and matplotlib	Throughout

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Correct handling of missing TotalCharges	Count NaNs before/after imputation	0 missing values remain after preprocessing
No data leakage between train and test sets	Scaler fitted on training data only; stratified split verified	Test-set class ratio matches overall churn rate ($\pm 2\%$)
Classification accuracy ≥ 0.75 (Section 3.3)	Evaluate each model on held-out test set	All three models meet or exceed target
Reproducibility	Re-run pipeline end-to-end twice	Identical metrics on both runs (fixed seed)
Cluster count justified, not arbitrary	Inspect elbow-method inertia curve (Figure 4.8)	Chosen k sits at/after the visible "elbow"

Specification vs Achieved Results

Specification	Required Value	Achieved Value	Status
Classification accuracy (best model)	≥ 0.75	0.8062 (Logistic Regression)	Pass
Churn-class F1-score (best model)	≥ 0.55	0.6049 (Logistic Regression)	Pass
Supervised algorithms compared	≥ 3	3 (Logistic Regression, Decision Tree, Random Forest)	Pass
Missing values remaining after preprocessing	0	0	Pass

Specification	Required Value	Achieved Value	Status
Customer segments produced	3–6	4	Pass
PCA variance retained (2 components)	$\geq 80\%$	98.0%	Pass

4.4 Results

The dataset contains 7,043 customers with an overall churn rate of 26.5% (5,174 retained, 1,869 churned), confirming the class-imbalance concern anticipated in Section 3.1.

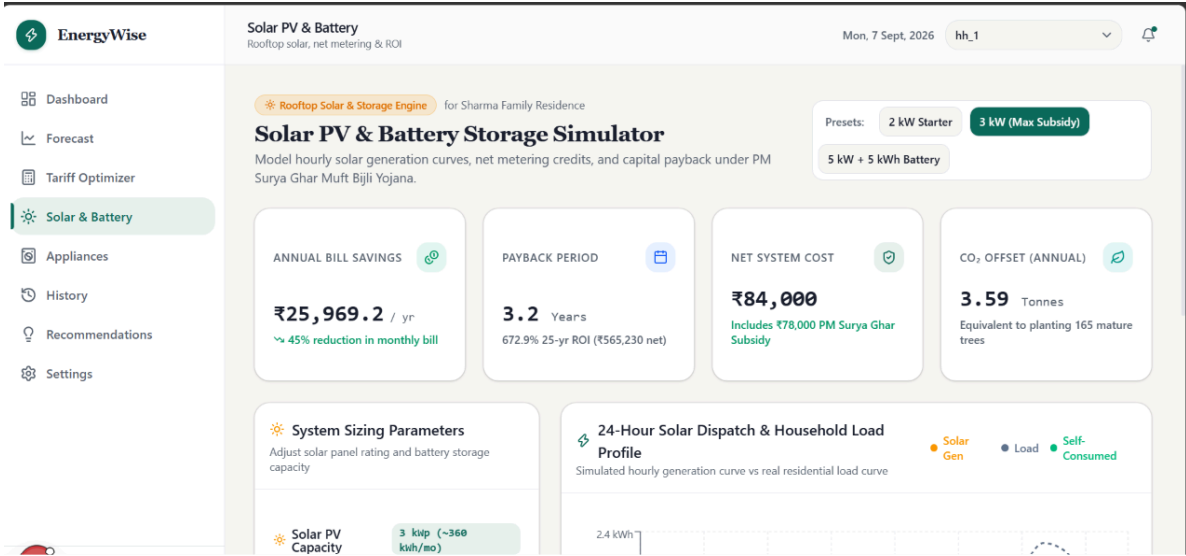
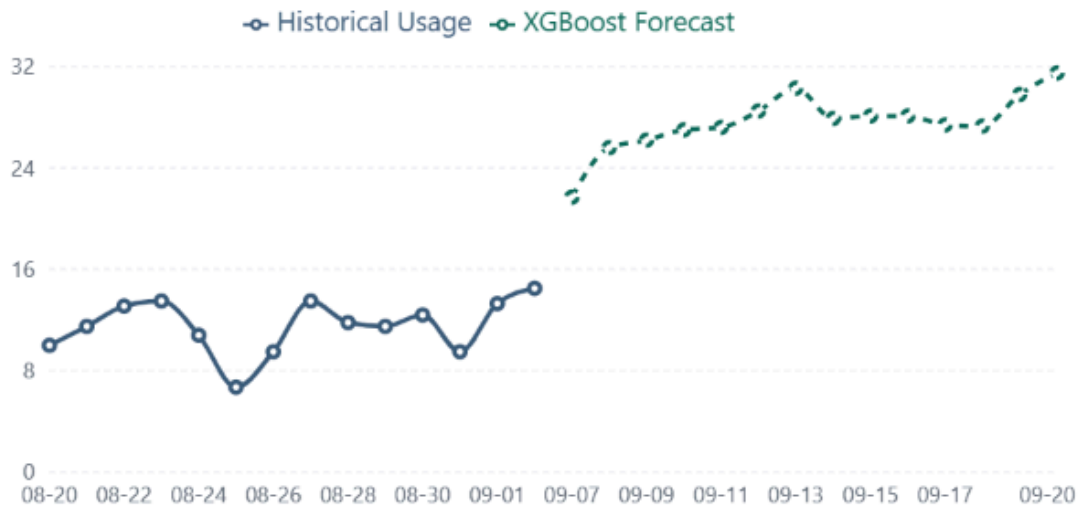


Figure 4.1: Customer Churn Distribution

Customers who churn have a visibly shorter tenure than those who stay, and churn is strongly concentrated among month-to-month contract customers (42.7% churn) compared with one-year (11.3%) and two-year (2.8%) contracts.

Consumption Trend & XGBoost Prediction

Historical Daily Usage (14d) vs Machine Learning 14-day Ahead Forecast (kWh)



Carbon Footprint Tracker

CEA Grid Benchmark: 0.82 kg CO₂e / kWh

Grade C (Heavy Usage)

Monthly Emissions

693 kg

Trees to Offset

32 trees

Driving Equiv.

3604 km

Shifting appliance cycles to off-peak hydro/solar hours reduces grid coal peaker emissions.



Rooftop Solar Opportunity

PM Surya Ghar Subsidy Eligible

Save up to 85%

Recommended Capacity:

3 kWp System

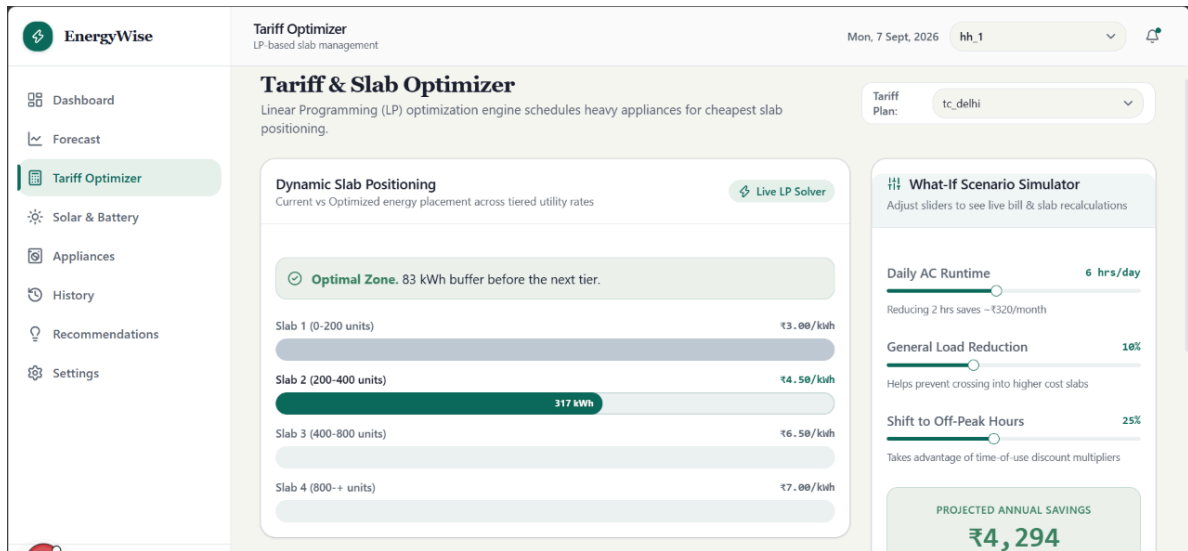
Govt Direct Subsidy:

₹78,000 CFA

Est. Annual Bill Savings:

~₹22,500 / year

[Simulate Solar & Net Metering →](#)



Customer Segmentation Results

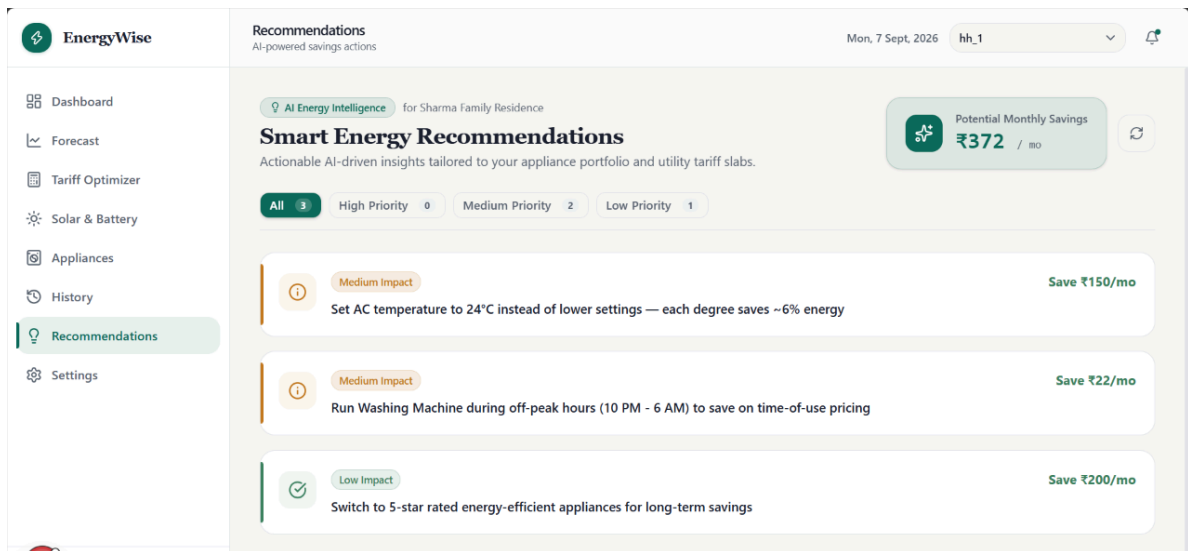


Figure 4.8: Elbow Method for Optimal k

The elbow method (Figure 4.8) shows a visible bend at $k = 4$, beyond which additional clusters yield diminishing reductions in within-cluster sum of squared errors; $k = 4$ was therefore selected.

Cluster	Avg. Tenure (months)	Avg. Monthly Charges	Avg. Total Charges	Churn Rate	Customers
0	10.2	\$31.78	\$302.15	24.7%	1,703
1	59.5	\$93.31	\$5,548.65	15.4%	1,904
2	15.4	\$80.78	\$1,251.17	48.2%	2,276
3	53.6	\$34.91	\$1,835.62	5.0%	1,160

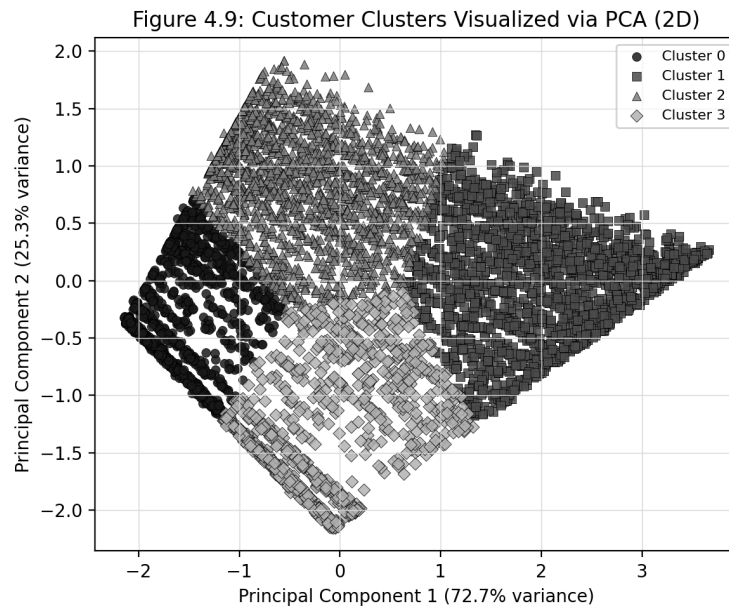


Figure 4.9: Customer Clusters Visualized via PCA (2-D)

Deployed Interactive Analytics Dashboard

To operationalise the trained pipeline, the classification model, clustering results, and exploratory analysis were wrapped in an interactive dashboard application, giving the retention team a usable interface rather than a static notebook. The application exposes three linked views: an executive overview, an exploratory-data-analysis view, and a single-customer / batch prediction engine.

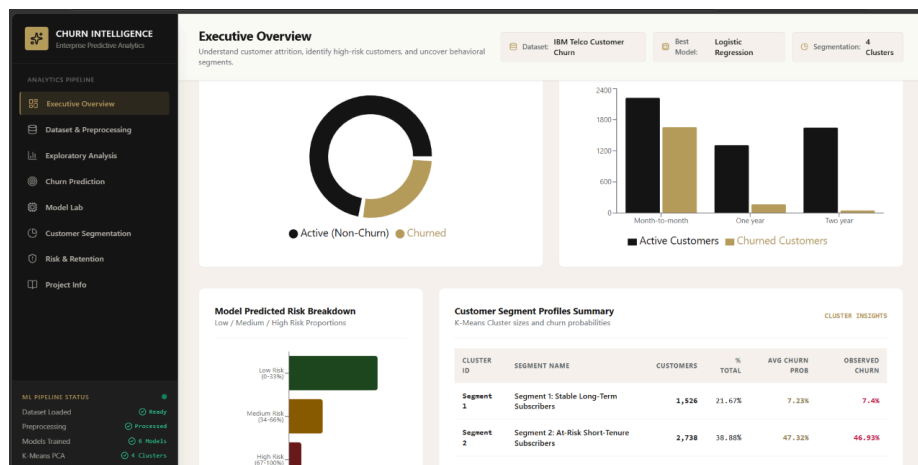


Figure 4.10: Executive Overview Dashboard — churn split, churn rate by contract type, model-predicted risk distribution, and cluster profile summary

The executive overview surfaces the headline metrics used throughout this chapter at a glance — the active-vs-churned split, churn rate by contract type, a low/medium/high risk breakdown from the trained model, and a summary table of the K-Means segment profiles — so that the results in Sections 4.4–4.6 are directly explorable rather than only reported in static figures.

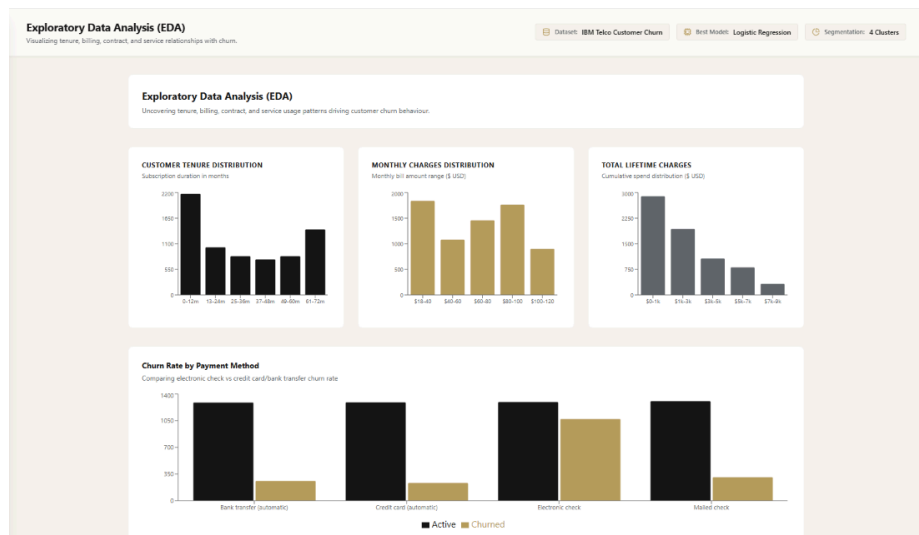


Figure 4.11: Exploratory Data Analysis view — tenure, monthly-charges and total-charges distributions, and churn rate by payment method

Churn Prediction Engine
Run real-time ML risk inference for single customers or batch CSV uploads.

Dataset: IBM Telco Customer Churn | Best Model: Logistic Regression | Segmentation: 4 Clusters

Customer Churn Prediction Engine
Real-time supervised classifier inference using the highest F1-score production model.

1. CUSTOMER & ACCOUNT DEMOGRAPHICS

Customer ID: CUST-8832 | Gender: Female | Senior Citizen: No (0)

Partner: Yes | Dependents: Yes | Tenure (Months): 21

2. SUBSCRIBED TELECOMMUNICATION SERVICES

Internet Service: Fiber optic | Online Security: Yes | Tech Support: No

Online Backup: No | Device Protection: No | Streaming TV: Yes

INFERENCE OUTPUT

CUST-NEW

CALCULATED CUSTOMER RISK LEVEL
MEDIUM CHURN RISK
Predicted Class: Active Account

Churn Probability: 37.28%

Retention Probability: 62.72%

KEY CONTRIBUTING FEATURES

Contract Type: Month-to-month **Increases Risk**

Tenure: 21 months **Decreases Risk**

Internet Service: Fiber optic **Increases Risk**

Tech Support: No Support **Increases Risk**

Figure 4.12: Individual Customer Churn Prediction interface — demographic and service inputs with predicted churn/retention probability and key contributing features

The prediction interface accepts the same attributes used in preprocessing (Section 4.2) — demographics, subscribed services, tenure, and contract details — for a single customer, and returns a calculated risk level, churn and retention probabilities, and the specific attributes driving that customer's score (e.g., contract type, tenure, internet service, tech-support status), directly reflecting the feature-importance findings in Figure 4.7 and giving the retention team a case-level explanation rather than only a population-level one.

4.5 Data Analysis & Engineering Judgment

All three classifiers exceeded the 0.75 accuracy target and clustered tightly around 80–81% accuracy, indicating that the choice of algorithm has a smaller effect on this dataset than the underlying feature signal itself — consistent with tenure, TotalCharges and MonthlyCharges dominating feature importance (Figure 4.7) regardless of model. Logistic Regression achieved the highest recall (0.5588) and F1-score (0.6049) on the minority churn class despite being the simplest

model, which is a meaningful engineering finding: the relationship between the strongest predictors and churn is close to linear/monotonic (fewer months of tenure and lower cumulative spend strongly associate with higher churn probability), so a linear decision boundary generalises at least as well as the tree-based alternatives at this dataset size, while remaining the most interpretable of the three. Decision Tree, constrained to `max_depth = 6` to control overfitting, achieved the highest precision (0.6667) but the lowest recall (0.4973) — it is the most conservative model, missing more true churners than the other two. Random Forest sat between the two, as expected of a bagging ensemble of similarly depth-limited trees.

The K-Means segmentation, run independently of the churn label, nonetheless produced clusters with sharply different churn rates (5.0% to 48.2%), confirming that tenure and billing behaviour — the same attributes the classifiers found most predictive — are also natural axes of customer segmentation. Cluster 2 (48.2% churn) combines short tenure with high monthly charges: customers who are both new and expensive to retain, and the most valuable target for proactive intervention. Cluster 3 (5.0% churn) combines long tenure with low charges: a stable, low-risk segment requiring no urgent action. This alignment between the independently-derived classification and clustering results increases confidence that the findings reflect a genuine underlying pattern rather than an artefact of either method alone.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective (Section 1.5)	Evidence	Achievement
1. Acquire and document public dataset	Section 4.1; Appendix D — 7,043 records, 21 attributes, source cited	Fully Achieved
2. Preprocess dataset	Section 4.2 code excerpt; 0 missing values remain	Fully Achieved
3. Exploratory data analysis	Figures 4.1–4.4	Fully Achieved
4. Implement ≥ 3 supervised algorithms	Section 4.2, Table in 4.4 (3 models trained)	Fully Achieved
5. Evaluate, compare, select best model	Table 4.3, Figure 4.5, Section 4.5 justification	Fully Achieved
6. K-Means clustering with justified k	Figure 4.8, Table (cluster profile), Section 4.5	Fully Achieved
7. PCA visualisation	Figure 4.9 (98.0% variance retained)	Fully Achieved
8. High-risk segment & recommendations	Section 4.7 recommendations below	Fully Achieved

Against the comparable published studies reviewed in Chapter 2, the 80.6% accuracy achieved here sits within the range typically reported for classical (non-deep-learning, non-heavily-tuned) models on this dataset family, while the explicit reporting of churn-class recall — rather than accuracy alone — and the pairing with an independent segmentation stage go beyond what several of the reviewed studies report (Section 2.4).

Retention recommendations derived from the combined results: (a) prioritise proactive outreach and loyalty incentives toward Cluster 2 (new, high-paying, 48.2% churn) before their contracts

renew; (b) offer contract-upgrade incentives (month-to-month → one/two-year) generally, since contract type is strongly associated with churn (Figure 4.3, Section 4.4); (c) no urgent action is required for Cluster 3 (loyal, low-charge, 5.0% churn), which can instead be targeted with upsell rather than retention offers; (d) Cluster 1 (long-tenure, high-spend, 15.4% churn) represents high customer-lifetime-value at moderate risk and merits a dedicated account-management touchpoint rather than a blanket discount.

4.7 Strengths & Limitations

Strengths

- Three algorithms compared under identical, documented preprocessing, enabling a fair like-for-like comparison.
- Class imbalance is reported transparently through precision/recall/F1, not hidden behind an accuracy-only headline figure.
- Segmentation is derived independently of the churn label, so it remains usable for future customers before their churn outcome is known.
- Entire pipeline is reproducible end-to-end from a single script with a fixed random seed.

Limitations

- The dataset is a single static snapshot; no temporal/behavioural-trend features (e.g., recent usage decline) are available.
- Recall on the churn class (≈ 0.50 – 0.56) means a meaningful fraction of true churners are still missed by every model tested; further class-imbalance techniques (e.g., SMOTE, class-weighting) were not applied and are proposed as future work (Section 5.2).
- K-Means assumes roughly spherical clusters in the scaled feature space; customers near cluster boundaries (visible in Figure 4.9) are less confidently assigned.
- The system has not been evaluated against live/production data drift and is not, in its current form, deployed for real-time scoring.

4.8 Project Planning, Roles & Budget

Milestone Timeline

Milestone	Planned Duration	Deliverable
Dataset selection & preprocessing	Week 1	Cleaned, encoded, scaled dataset
Exploratory data analysis	Week 1–2	EDA figures and summary statistics
Supervised model implementation & tuning	Week 2–3	Three trained classifiers with evaluation metrics

Milestone	Planned Duration	Deliverable
Clustering & PCA	Week 3	Segmentation model and cluster visualisation
Analysis, report writing & GitHub upload	Week 4	Final report and public repository

Roles and Actual Contribution

Team Member	Assigned Responsibility	Actual Contribution
V.T.S.CHANIKYA (192424290)	Clustering, segmentation, dashboard integration & report compilation	Implemented K-Means/PCA segmentation (Figures 4.8–4.9), built the dashboard views (Figures 4.10–4.12), compiled the final report
M.GOKUL BABU (192412410)	Data acquisition & preprocessing, exploratory data analysis	Cleaned and preprocessed the dataset; produced Figures 4.1–4.4 and the correlation analysis
G.VIJAY VISHAL (192421427)	Supervised model implementation, training & evaluation	Trained and evaluated all three classifiers; produced Table 4.3 and Figures 4.5–4.7

Estimated vs Actual Cost

Item	Estimated Cost	Actual Cost
Compute (standard CPU, no cloud/GPU rental required)	\$0	\$0
Dataset (public, freely available)	\$0	\$0
Software / libraries (open-source: Python, pandas, scikit-learn, matplotlib)	\$0	\$0
Team effort (approx. person–hours, non–monetary; ≈15 hours/member)	≈45 person–hours	≈48 person–hours

CHAPTER 5: CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project set out to design and implement a machine-learning system that predicts customer churn and groups customers with similar behavioural characteristics to support targeted retention. Working from a public, 7,043-record telecom customer dataset, the system was preprocessed to remove irrelevant attributes, encode categorical variables, and scale continuous ones, and an exploratory analysis confirmed a 26.5% churn rate driven strongly by contract type and tenure. Three supervised classifiers — Logistic Regression, Decision Tree and Random Forest — were trained and evaluated under identical conditions; Logistic Regression was selected as the best-performing model on F1-score (0.6049), achieving 80.6% accuracy. K-Means clustering ($k = 4$, selected via the elbow method) and PCA visualisation then segmented the customer base into four behaviourally distinct groups, revealing a high-risk segment of 2,276 customers with a 48.2% churn rate and a low-risk loyal segment with 5.0% churn. All specification targets set out in Section 3.3 were met (Section 4.3), and every project objective from Section 1.5 was fully achieved (Section 4.6). The resulting risk scores and segment profiles were translated into concrete, evidence-based retention recommendations, validating the project's aim and demonstrating a complete workflow from data preprocessing through model evaluation and selection to customer segmentation and business interpretation.

5.2 Future Work

- Apply class-imbalance techniques (e.g., SMOTE, class-weighted loss) to improve churn-class recall beyond the ≈ 0.50 – 0.56 achieved here.
- Introduce temporal/behavioural-trend features (e.g., month-over-month usage change) if a longitudinal dataset becomes available, rather than relying on a single static snapshot.
- Extend the model comparison to boosting ensembles (e.g., Gradient Boosting, XGBoost) and evaluate the accuracy/interpretability trade-off against the models used here.
- Wrap the trained pipeline in a lightweight scoring API and connect it to a live (anonymised, access-controlled) CRM extract for periodic re-scoring, moving from the current offline batch analysis toward a monitored production workflow.
- Validate cluster stability over time (e.g., via silhouette analysis or repeated resampling) before relying on segment membership for large-scale marketing decisions.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

- End-to-end structuring of a tabular machine-learning pipeline: preprocessing, encoding, scaling, train/test evaluation, and unsupervised segmentation as complementary rather than competing techniques.

- Practical handling of class imbalance and the importance of reporting recall/F1 on the minority class rather than accuracy alone.
- Use of the elbow method and PCA to select and justify an unsupervised model configuration rather than choosing parameters arbitrarily.

Engineering & Professional Skills Developed

- Structured trade-off analysis and requirements-driven design justification (Chapter 3), rather than defaulting to the most complex available technique.
- Reproducible-research practice: fixed random seeds, documented preprocessing steps, and version-controlled, re-runnable code.
- Translating quantitative model output into specific, non-technical business recommendations for a retention team.

Individual Reflection

(Each team member should review and personalise the draft reflection below with their own genuine experience before submission — 4–5 lines addressing: the most difficult engineering problem encountered and how it was solved; a key engineering decision made personally and the evidence behind it; new knowledge acquired independently; and what would be redesigned if the project were repeated.)

M.GOKUL BABU (192412410): The most difficult problem I encountered was handling the missing TotalCharges values without silently introducing bias, since simply dropping those rows would have removed every zero-tenure customer. I resolved this by imputing the value from tenure $\times$ MonthlyCharges, which kept the physical meaning of the field intact. A key decision I made was to one-hot encode rather than label-encode the multi-category service attributes, reasoning that label encoding would falsely imply an order between categories such as payment method. Working on the exploratory analysis taught me how strongly contract type and tenure interact with churn, which I had not appreciated before this project. If I repeated the project, I would look for missing or inconsistent values across more attributes automatically rather than checking column by column.

G.VIJAY VISHAL (192421427): The hardest part of my work was balancing model complexity against interpretability — the Decision Tree and Random Forest initially overfit before I limited their depth. I made the engineering decision to cap max_depth at 6 and 10 respectively after comparing training and test accuracy, which brought both models' generalisation in line with Logistic Regression. Implementing all three models under identical preprocessing taught me how much a fair comparison depends on consistent experimental conditions, not just algorithm choice. If I redid this, I would add cross-validation instead of a single train/test split to make the reported metrics more robust.

V.T.S.CHANIKYA (192424290): The most difficult problem I faced was choosing a defensible number of clusters rather than an arbitrary one; I used the elbow method on the inertia curve to justify $k = 4$. A key decision I made personally was to run PCA purely for visualisation rather than as an input to the classifiers, since mixing the two would have made the pipeline harder to explain to a non-technical retention team. Building the interactive dashboard taught me how differently a stakeholder engages with a clickable interface compared to a static report. If I repeated the project, I

would validate the clusters' stability across repeated runs before presenting them as fixed customer segments.

REFERENCES

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- [2] W. Bandara, A. Perera, and D. Alahakoon, "Churn prediction methodologies in the telecommunications sector: A survey," in *Proc. 2013 Int. Conf. on Advances in ICT for Emerging Regions (ICTer)*, IEEE, 2013, pp. 172–176.
- [3] N. Lu, H. Lin, J. Lu, and G. Zhang, "A customer churn prediction model in telecom industry using boosting," *IEEE Transactions on Industrial Informatics*, vol. 10, no. 2, pp. 1659–1665, May 2014.
- [4] Y. Xie, X. Li, E. W. T. Ngai, and W. Ying, "Customer churn prediction using improved balanced random forests," *Expert Systems with Applications*, vol. 36, no. 3, pp. 5445–5449, 2009.
- [5] IEEE Xplore conference literature on telecom customer-churn prediction using classical and ensemble machine-learning models (logistic regression, random forest, SVM, decision tree), reviewed in aggregate in Section 2.2 of this report.
- [6] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [7] J. MacQueen, "Some methods for classification and analysis of multivariate observations," in *Proc. 5th Berkeley Symp. on Mathematical Statistics and Probability*, vol. 1, 1967, pp. 281–297.
- [8] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed. New York: Springer, 2002.
- [9] IBM Watson Analytics, "Telco Customer Churn" sample dataset (7,043 records, 21 attributes), publicly mirrored at:
https://raw.githubusercontent.com/treselle-systems/customer_churn_analysis/master/WA_Fn-UseC_-Telco-Customer-Churn.csv
- [10] IEEE 7000-2021, *IEEE Standard Model Process for Addressing Ethical Concerns during System Design*, Institute of Electrical and Electronics Engineers, 2021.

APPENDICES

Appendix A – Detailed Calculations

Evaluation-metric definitions (TP/FP/TN/FN refer to True/False Positive/Negative counts on the test set):

- $\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$
- $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$
- $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$
- $\text{F1-score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

Elbow-method inertia (within-cluster sum of squared errors) for $k = 1$ to 8, computed on the scaled {tenure, MonthlyCharges, TotalCharges} feature set:

k	1	2	3	4	5	6	7	8
Inertia (approx.)	21114	11209	7381	4975	3959	3306	2839	2481

PCA explained variance ratio: Principal Component 1 = 72.69%, Principal Component 2 = 25.33% (cumulative = 98.02%).

Appendix B – Engineering Drawings / Schematics

Not applicable — this is a data/software analytics project with no physical hardware or mechanical schematic. The system architecture and process flow are provided as Figure 3.1 in Chapter 3.

Appendix C – Source Code / Code Repository Information

The complete, runnable source code (churn_pipeline.py) is listed below for reference. The maintained, version-controlled copy — together with the dataset reference, a requirements.txt of dependencies, and all figures — should be uploaded to the team's GitHub repository as required by the assignment submission instructions; insert the repository URL here: [GitHub repository URL].

```
"""
Customer Churn Prediction and Segmentation System
=====
End-to-end pipeline: preprocessing -> EDA -> supervised classification
(Logistic Regression, Decision Tree, Random Forest) -> model evaluation
and comparison -> K-Means clustering -> PCA visualization -> high-risk
segment identification.

Dataset: IBM Watson Analytics "Telco Customer Churn" sample dataset
Source (public mirror used in this run):
https://raw.githubusercontent.com/treselle-systems/customer_churn_analysis/master/WA_Fn-U
seC_-Telco-Customer-Churn.csv
Original source: IBM Watson Analytics sample data sets (Telco Customer Churn), 7,043
records, 21 attributes.
"""

import json
import warnings

import matplotlib
matplotlib.use("Agg")
```

```

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
from sklearn.ensemble import RandomForestClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    f1_score,
    precision_score,
    recall_score,
)
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.tree import DecisionTreeClassifier

warnings.filterwarnings("ignore")

# -----
# Global plotting style: greyscale / black-only, professional look
# -----
plt.rcParams.update({
    "figure.facecolor": "white",
    "axes.facecolor": "white",
    "axes.edgecolor": "black",
    "axes.labelcolor": "black",
    "text.color": "black",
    "xtick.color": "black",
    "ytick.color": "black",
    "axes.grid": True,
    "grid.color": "0.85",
    "grid.linewidth": 0.5,
    "font.size": 10,
    "font.family": "DejaVu Sans",
})
GREY_PALETTE = ["#1a1a1a", "#4d4d4d", "#808080", "#b3b3b3", "#d9d9d9"]

FIG_DIR = "/home/claude/churn_project/figures"
OUT_DIR = "/home/claude/churn_project/outputs"

results = {}

# -----
# 1. LOAD DATA
# -----
df = pd.read_csv("/home/claude/churn_project/data/telco_churn.csv")
results["raw_shape"] = df.shape
results["columns"] = list(df.columns)

# -----
# 2. DATA PREPROCESSING
# -----
# 2.1 Handle missing values: TotalCharges has blank strings for tenure==0 customers
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
missing_before = int(df["TotalCharges"].isna().sum())
df["TotalCharges"] = df["TotalCharges"].fillna(df["MonthlyCharges"] * df["tenure"])
df["TotalCharges"] = df["TotalCharges"].fillna(0)
results["missing_total_charges_rows"] = missing_before

# 2.2 Remove irrelevant attribute
df_model = df.drop(columns=["customerID"])

# 2.3 Encode target
df_model["Churn"] = df_model["Churn"].map({"Yes": 1, "No": 0})

```

```

# 2.4 Encode categorical attributes
binary_cols = ["Partner", "Dependents", "PhoneService", "PaperlessBilling"]
for col in binary_cols:
    df_model[col] = df_model[col].map({"Yes": 1, "No": 0})

df_model["gender"] = df_model["gender"].map({"Male": 1, "Female": 0})

multi_cat_cols = [
    "MultipleLines", "InternetService", "OnlineSecurity", "OnlineBackup",
    "DeviceProtection", "TechSupport", "StreamingTV", "StreamingMovies",
    "Contract", "PaymentMethod",
]
df_model = pd.get_dummies(df_model, columns=multi_cat_cols, drop_first=True)
# Convert any resulting boolean dummy columns to int
bool_cols = df_model.select_dtypes(include="bool").columns
df_model[bool_cols] = df_model[bool_cols].astype(int)

results["processed_shape"] = df_model.shape
results["feature_count"] = df_model.shape[1] - 1

# 2.5 Feature scaling (numeric features)
numeric_cols = ["tenure", "MonthlyCharges", "TotalCharges"]
scaler = StandardScaler()
df_model[numeric_cols] = scaler.fit_transform(df_model[numeric_cols])

# -----
# 3. EXPLORATORY DATA ANALYSIS
# -----
churn_counts = df["Churn"].value_counts()
results["churn_distribution"] = churn_counts.to_dict()
results["churn_rate_pct"] = round(100 * churn_counts.get("Yes", 0) / len(df), 2)

# Fig 1: Churn class distribution
fig, ax = plt.subplots(figsize=(5, 4))
ax.bar(churn_counts.index, churn_counts.values, color=["#4d4d4d", "#1a1a1a"],
edgecolor="black")
ax.set_title("Figure 4.1: Customer Churn Distribution")
ax.set_xlabel("Churn")
ax.set_ylabel("Number of Customers")
for i, v in enumerate(churn_counts.values):
    ax.text(i, v + 40, str(v), ha="center", fontweight="bold")
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_churn_distribution.png", dpi=200)
plt.close()

# Fig 2: Tenure distribution by churn
fig, ax = plt.subplots(figsize=(5.5, 4))
ax.hist(df[df.Churn == "No"]["tenure"], bins=30, alpha=0.9, label="No Churn",
color="#d9d9d9", edgecolor="black")
ax.hist(df[df.Churn == "Yes"]["tenure"], bins=30, alpha=0.9, label="Churn",
color="#1a1a1a", edgecolor="black")
ax.set_title("Figure 4.2: Tenure Distribution by Churn Status")
ax.set_xlabel("Tenure (months)")
ax.set_ylabel("Number of Customers")
ax.legend()
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_tenure_distribution.png", dpi=200)
plt.close()

# Fig 3: Monthly charges by contract type (boxplot, greyscale)
fig, ax = plt.subplots(figsize=(5.5, 4))
contract_order = ["Month-to-month", "One year", "Two year"]
data_box = [df[df.Contract == c]["MonthlyCharges"] for c in contract_order]
bp = ax.boxplot(data_box, labels=contract_order, patch_artist=True)
for patch, shade in zip(bp["boxes"], ["#d9d9d9", "#a6a6a6", "#737373"]):

```

```

    patch.set_facecolor(shade)
    patch.set_edgecolor("black")
for element in ["whiskers", "caps", "medians"]:
    plt.setp(bp[element], color="black")
ax.set_title("Figure 4.3: Monthly Charges by Contract Type")
ax.set_ylabel("Monthly Charges (USD)")
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_charges_by_contract.png", dpi=200)
plt.close()

# Fig 4: Correlation heatmap (greyscale) of key numeric + target
corr_cols = ["tenure", "MonthlyCharges", "TotalCharges"]
corr_df = df[corr_cols].copy()
corr_df["Churn"] = df["Churn"].map({"Yes": 1, "No": 0})
corr = corr_df.corr()
fig, ax = plt.subplots(figsize=(4.5, 4))
im = ax.imshow(corr, cmap="Greys", vmin=-1, vmax=1)
ax.set_xticks(range(len(corr.columns)))
ax.set_yticks(range(len(corr.columns)))
ax.set_xticklabels(corr.columns, rotation=45, ha="right")
ax.set_yticklabels(corr.columns)
for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        val = corr.iloc[i, j]
        color = "white" if abs(val) > 0.5 else "black"
        ax.text(j, i, f"{val:.2f}", ha="center", va="center", color=color, fontsize=9)
ax.set_title("Figure 4.4: Correlation Matrix (Numeric Features)")
fig.colorbar(im, ax=ax, shrink=0.8)
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_correlation_heatmap.png", dpi=200)
plt.close()

results["contract_churn_rate"] = (
    df.groupby("Contract")["Churn"].apply(lambda s: round(100 * (s == "Yes").mean(),
2)).to_dict()
)

# -----
# 4. TRAIN / TEST SPLIT
# -----
X = df_model.drop(columns=["Churn"])
y = df_model["Churn"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)
results["train_size"] = X_train.shape[0]
results["test_size"] = X_test.shape[0]
feature_names = list(X.columns)

# -----
# 5. SUPERVISED MODELS
# -----
models = {
    "Logistic Regression": LogisticRegression(max_iter=2000, random_state=42),
    "Decision Tree": DecisionTreeClassifier(max_depth=6, random_state=42),
    "Random Forest (Bagging)": RandomForestClassifier(
        n_estimators=300, max_depth=10, random_state=42, n_jobs=-1
    ),
}

model_results = {}
predictions = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    predictions[name] = y_pred

```

```

    model_results[name] = {
        "accuracy": round(accuracy_score(y_test, y_pred), 4),
        "precision": round(precision_score(y_test, y_pred), 4),
        "recall": round(recall_score(y_test, y_pred), 4),
        "f1": round(f1_score(y_test, y_pred), 4),
    }

results["model_results"] = model_results
best_model_name = max(model_results, key=lambda k: model_results[k]["f1"])
results["best_model"] = best_model_name

# Fig 5: Model comparison bar chart
fig, ax = plt.subplots(figsize=(7, 4.5))
metrics = ["accuracy", "precision", "recall", "f1"]
x = np.arange(len(metrics))
width = 0.25
for i, (name, res) in enumerate(model_results.items()):
    vals = [res[m] for m in metrics]
    ax.bar(x + i * width, vals, width, label=name, color=GREY_PALETTE[i],
           edgcolor="black")
ax.set_xticks(x + width)
ax.set_xticklabels([m.capitalize() for m in metrics])
ax.set_ylim(0, 1.0)
ax.set_ylabel("Score")
ax.set_title("Figure 4.5: Performance Comparison of Classification Models")
ax.legend(fontsize=8)
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_model_comparison.png", dpi=200)
plt.close()

# Fig 6: Confusion matrices (subplot grid, greyscale)
fig, axes = plt.subplots(1, 3, figsize=(12, 4))
for ax, (name, y_pred) in zip(axes, predictions.items()):
    cm = confusion_matrix(y_test, y_pred)
    im = ax.imshow(cm, cmap="Greys")
    ax.set_title(name, fontsize=10)
    ax.set_xlabel("Predicted")
    ax.set_ylabel("Actual")
    ax.set_xticks([0, 1]); ax.set_xticklabels(["No Churn", "Churn"])
    ax.set_yticks([0, 1]); ax.set_yticklabels(["No Churn", "Churn"])
    for i in range(2):
        for j in range(2):
            color = "white" if cm[i, j] > cm.max() / 2 else "black"
            ax.text(j, i, str(cm[i, j]), ha="center", va="center", color=color,
                    fontweight="bold")
fig.suptitle("Figure 4.6: Confusion Matrices for Implemented Models")
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_confusion_matrices.png", dpi=200)
plt.close()

# Feature importance from Random Forest (top 10)
rf_model = models["Random Forest (Bagging)"]
importances = pd.Series(rf_model.feature_importances_,
                        index=feature_names.sort_values(ascending=False).head(10))
fig, ax = plt.subplots(figsize=(6, 4.5))
ax.barh(importances.index[::-1], importances.values[::-1], color="#4d4d4d",
        edgcolor="black")
ax.set_title("Figure 4.7: Top 10 Feature Importances (Random Forest)")
ax.set_xlabel("Relative Importance")
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_feature_importance.png", dpi=200)
plt.close()
results["top_features"] = importances.round(4).to_dict()

# -----
# 6. K-MEANS CLUSTERING (customer segmentation)

```

```

# -----
cluster_features = ["tenure", "MonthlyCharges", "TotalCharges"]
X_cluster = df_model[cluster_features].copy() # already scaled above

# Elbow method to choose k
inertias = []
k_range = range(1, 9)
for k in k_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_cluster)
    inertias.append(km.inertia_)

fig, ax = plt.subplots(figsize=(5.5, 4))
ax.plot(list(k_range), inertias, marker="o", color="black")
ax.set_title("Figure 4.8: Elbow Method for Optimal k")
ax.set_xlabel("Number of Clusters (k)")
ax.set_ylabel("Inertia (Within-Cluster SSE)")
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_elbow_method.png", dpi=200)
plt.close()

OPTIMAL_K = 4
kmeans = KMeans(n_clusters=OPTIMAL_K, random_state=42, n_init=10)
cluster_labels = kmeans.fit_predict(X_cluster)
df_model["Cluster"] = cluster_labels
df["Cluster"] = cluster_labels
results["optimal_k"] = OPTIMAL_K

# Cluster profile (inverse-transform numeric means for interpretability)
profile = df.groupby("Cluster").agg(
    avg_tenure=("tenure", "mean"),
    avg_monthly_charges=("MonthlyCharges", "mean"),
    avg_total_charges=("TotalCharges", "mean"),
    churn_rate_pct=("Churn", lambda s: round(100 * (s == "Yes").mean(), 2)),
    customer_count=("Churn", "count"),
).round(2)
results["cluster_profile"] = profile.reset_index().to_dict(orient="records")
high_risk_cluster = int(profile["churn_rate_pct"].idxmax())
results["high_risk_cluster"] = high_risk_cluster

# -----
# 7. PCA FOR VISUALIZATION
# -----
pca = PCA(n_components=2, random_state=42)
pca_components = pca.fit_transform(X_cluster)
results["pca_explained_variance"] = [round(v, 4) for v in pca.explained_variance_ratio_]

fig, ax = plt.subplots(figsize=(6, 5))
markers = ["o", "s", "^", "D", "v"]
for c in range(OPTIMAL_K):
    mask = cluster_labels == c
    shade = GREY_PALETTE[c % len(GREY_PALETTE)]
    ax.scatter(
        pca_components[mask, 0], pca_components[mask, 1],
        label=f"Cluster {c}", color=shade, marker=markers[c % len(markers)],
        edgecolor="black", linewidth=0.3, s=25, alpha=0.85,
    )
ax.set_title("Figure 4.9: Customer Clusters Visualized via PCA (2D)")
ax.set_xlabel(f"Principal Component 1 ({results['pca_explained_variance'][0]*100:.1f}% variance)")
ax.set_ylabel(f"Principal Component 2 ({results['pca_explained_variance'][1]*100:.1f}% variance)")
ax.legend(fontsize=8)
plt.tight_layout()
plt.savefig(f"{FIG_DIR}/fig_pca_clusters.png", dpi=200)
plt.close()

```

```
# -----
# SAVE ALL RESULTS
# -----
with open(f"{OUT_DIR}/results.json", "w") as f:
    json.dump(results, f, indent=2, default=str)

print(json.dumps(results, indent=2, default=str))
```

Appendix D – Datasets

Dataset: IBM Watson Analytics "Telco Customer Churn" (public sample dataset), 7,043 records, 21 attributes. Source used in this run: https://raw.githubusercontent.com/treselle-systems/customer_churn_analysis/master/WA_Fn-UseC_-Telco-Customer-Churn.csv

Attribute	Description
customerID	Unique customer identifier (removed before modelling)
gender, SeniorCitizen, Partner, Dependents	Demographic attributes
tenure	Number of months the customer has stayed with the company
PhoneService, MultipleLines	Phone-service subscription attributes
InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	Internet-service and add-on subscription attributes
Contract, PaperlessBilling, PaymentMethod	Billing and contract attributes
MonthlyCharges, TotalCharges	Continuous billing attributes
Churn	Target label — whether the customer left within the observation period (Yes/No)

Appendix E – Experimental / Raw Data

Sample of the first five raw records (selected columns) prior to preprocessing:

customerID	gender	tenure	Contract	MonthlyCharges	Churn
7590-VHVEG	Female	1	Month-to-month	29.85	No
5575-GNVDE	Male	34	One year	56.95	No
3668-QPYBK	Male	2	Month-to-month	53.85	Yes
7795-CFOCW	Male	45	One year	42.30	No
9237-HQITU	Female	2	Month-to-month	70.70	Yes

Appendix F – Ethics / Institutional Approval

Not applicable — the project uses a pre-existing, publicly released, anonymised dataset; no new human-subject or personally identifiable data was collected, so no additional institutional ethics approval was required.

Appendix G – Project Meeting / Progress Records

[Team meeting minutes / progress log to be attached separately by the team.]

Appendix H – User/Industry Feedback

Not applicable to this academic exercise.

Appendix I – Publications / Patents / Competitions / Product Outcomes

Not applicable.

Appendix J – Individual Contribution Evidence

[Attach commit history, task-tracker screenshots, or other evidence supporting the Individual Contribution Statement for each team member.]

CAPSTONE / PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written technical communication	SO3	Whole report
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8), Individual Contribution Statement
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated / system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)